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PERFORMANCE SPECIFICATION

CONTROLLER, PROGRAMMABLE, NAVAL SHIPBOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three sizes and capabilities of high-end programmable logic controllers (PLC) (see 6.4.10) requiring cybersecurity measures for naval shipboard use. The configuration range includes rack-mounted, rack-less, and module-type PLCs. The terms “PLC” and “programmable controller” are used interchangeably throughout this specification.

1.2 Classification. PLCs are of the following types:

a. Compact PLC: An integrated, compact unit that is mounted or placed beside the equipment to be controlled. The processor, power supply, and input/output (I/O) points are integrated within a single enclosure with I/O expansion capability.

b. Modular PLC: A programmable controller built with several components that are connected into a common rack or bus containing a processor, I/O modules (see 6.4.7), and power supply.

(1) Small Modular PLC: A unit that is mounted or placed beside the equipment to be controlled with the capability to control a single system.

(2) Large Modular PLC: A unit designed to control multiple independent systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment
MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ELECTROTECHNICAL COMMISSION

IEC 60715	-	Dimensions of Low-Voltage Switchgear and Controlgear – Standardized Mounting on Rails for Mechanical Support of Switchgear, Controlgear and Accessories
IEC 61131-1	-	Programmable Controllers – Part 1: General Information
IEC 61131-2	-	Programmable Controllers – Part 2: Equipment Requirements and Tests
IEC 61131-3	-	Programmable Controllers – Part 3: Programming Languages
IEC 61131-5	-	Programmable Controllers – Part 5: Communications
IEC 62443-4-1	-	Security for Industrial Automation and Control Systems – Part 4-1: Secure Product Development Lifecycle Requirements

(Copies of these documents are available online at http://webstore.iec.ch.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Design and components.

3.2.1 Included parts.

a. The Compact programmable controller shall be supplied with a central processing unit (CPU), integrated memory, integrated inputs and outputs, power supply, and the ability to be panel-mounted and DIN-mounted (in accordance with IEC 60715).

b. The Small Modular programmable controller shall be supplied with a CPU, integrated memory, network interface(s), power supply, and the ability to be panel-mounted and DIN-mounted (in accordance with IEC 60715).

c. The Large Modular programmable controller shall be supplied with a CPU, integrated memory, network interface(s), power supply, and the ability to be panel-mounted.

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3.2.2 Design. The Compact, Small Modular, and Large Modular programmable controller shall be designed to allow an external interface to existing systems. The processor and input and output circuitry shall have a modular, field-expandable design. The PLC design shall allow for expansion of the unit by the addition of hardware and user software.

3.2.3 Component compatibility. The Large Modular programmable controller shall have downward compatibility whereby a newly designed module can be interchanged with a previously designed module to reduce obsolescence or redesign of the system being controlled. The PLC unit shall be distributed and scalable with a reconfigurable architecture.

3.2.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Cybersecurity measures. Secure product development practices shall be implemented for the development of all programmable controllers in accordance with IEC 62443-4-1. At a minimum, Compact, Small Modular, and Large Modular programmable controllers and their associated circuitry or programming software shall have the following cybersecurity measures implemented.

a. All programmable controllers shall include access protections to prevent actions from malicious data on the removable memory.

b. All programmable controllers shall have the capability to generate an alert to the supervisory network (see 6.4.12) and locally for standalone programmable controllers upon exceeding a programmable parameter's operating range.

c. All programmable controllers shall have the capability to configure memory addresses to be read/write, read-only, or no access by an external component.

d. Small Modular and Large Modular programmable controller programming software shall have the ability to allow administrator access, enable security features, and create, modify, disable, and remove user accounts.

(1) Each account shall have rights in accordance with the principle of least privilege.

(2) Factory default authentication credentials shall be removed or changed prior to delivery or installation.

(3) All programmable controllers shall have the ability to configure the complexity of passwords of a minimum of fifteen characters, including upper case, lower case, numeric, and special characters.

(4) All programmable controllers shall have the ability to log user log-in, log-out, and access for any function.

e. All programmable controllers' wireless networking functions shall have the ability to be disabled.

f. All programmable controllers shall have the ability to implement firewall rules to include communication services, protocols, and access.

g. All programmable controllers shall have the ability to monitor I/O configuration and firmware and prevent unauthorized use of unwanted modules and firmware.

h. All programmable controllers shall have the ability to monitor and provide indication of unauthorized changes to the configuration of the PLC.

i. Small Modular and Large Modular programmable controllers shall have the ability to prevent access to manipulate memory addresses by the controller.

j. All unused or unnecessary programmable controller ports shall have the ability of being disabled logically or locked physically.

k. All programmable controllers shall have the capability of disabling or removing unwanted built-in functionality, such as a built-in web browser.

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l. The Small Modular and Large Modular programmable controller shall have the capability for unused slots to be disabled logically or locked physically.

m. The Compact, Small Modular, and Large Modular programmable controller shall have the ability to store application code within write-protected recoverable memory.

3.4 Performance characteristics.

3.4.1 Reliability. Programmable controller components shall be designed such that the mean time between failures is a minimum of 100,000 hours of operation, where failure is any condition that prevents the controller from meeting any requirements herein.

3.4.2 Maintainability. The mean time to repair shall be not greater than 3 hours, and the maximum time to repair shall be not greater than 6 hours. The mean maintenance time per month shall be not greater than 30 minutes. The Large Modular programmable controller shall have the ability to remove an I/O module from the programmable controller chassis, while power is applied, without faulting the controller or damaging the I/O module.

3.4.3 Duty. The Compact, Small Modular, and Large Modular programmable controller shall be constructed to operate for continuous duty (see 6.4.2) and general-purpose service (see 6.4.11) in accordance with MIL-DTL-2212.

3.4.4 Diagnostics. The Compact, Small Modular, and Large Modular programmable controller shall be in accordance with IEC 61131-2 and shall have the following diagnostic measures:

a. Diagnostic information for module and component failures

b. Processor scan times

c. Incorrect module status

d. Missing module status

e. Module failed status

f. Communication link loss status

g. Communication message failure diagnostics

h. Message alarming on programmable controller status

i. The Large Modular programmable controller shall have the ability to monitor the status of and perform diagnostic checks on other programmable controllers on the network.

3.4.5 CPU.

3.4.5.1 Function. The Compact, Small Modular, and Large Modular programmable controller shall provide control program execution, input and output scanning, and user program uploads and downloads over the network.

3.4.5.2 Firmware. The Compact, Small Modular, and Large Modular PLC operating system shall be contained in non-volatile memory and shall be upgradeable with the following features:

a. The programmable controller shall be protected from unauthorized alteration of non-volatile memory. Methods shall be provided to prevent unauthorized changes to firmware.

b. Firmware running on the programmable controller shall be digitally signed by the original equipment manufacturer using a trusted Root Certificate Authority.

3.4.5.3 Memory. The Compact, Small Modular, and Large Modular programmable controller shall be provided with base processor memory for user program and data storage as specified (see 6.2).

a. The Compact programmable controller shall contain a minimum of 10 kilobytes (KB) of integrated work memory.

b. The Small Modular programmable controller shall contain a minimum of 1 megabyte (MB) of integrated work memory.

c. The Large Modular programmable controller shall contain a minimum of 2 MB of integrated work memory.

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d. The Compact, Small Modular, and Large Modular programmable controller operating system, user program, and data shall be contained in random access memory (RAM)-type memory with the ability to be backed up with removable media.

e. The Compact, Small Modular, and Large Modular programmable controller shall have a physical slot for a removable memory module or card to be connected to the PLC for application-specific program file back-up, transport, and storage to retain data in the event of corruption or power loss.

f. All user memory in the Compact, Small Modular, and Large Modular programmable controller not used for program storage shall be allocable from the main memory for the purpose of data storage. The programmable controller shall be capable of storing the following predefined and user-defined data types:

(1) Boolean values

(2) Control structure

(3) Counter values

(4) Integer values

(5) Message values

(6) Real numbers

(7) Timer values

(8) External output status

(9) External input status

(10) Decimal numbers

(11) Binary numbers

(12) Binary coded decimal (BCD) numbers

(13) Internal processor status information

(14) ASCII character data

(15) ASCII string data

3.4.5.4 Operation mode. The Small Modular and Large Modular programmable controller operation mode shall have the ability to be changed within the programming software.

3.4.5.5 Processor fault indication. The Compact, Small Modular, and Large Modular programmable controller shall:

a. Have the ability to provide a fault status.

b. Contain visual indication of a fault on the programmable controller's front panel. This may be a light, such as a light-emitting diode (LED) dedicated to fault status, or an alternate blinking pattern to alert the user of a present fault.

c. Provide the ability to view and diagnose programmable controller faults in the programming software for troubleshooting and to be cleared/reset.

d. Upon fault indication, have the ability to configure the I/Os' failure state to be fail low (see 6.4.5), fail high (see 6.4.3), or fail last (see 6.4.4).

e. Have the capability to create a fault recovery routine that will identify and handle major faults. The fault recovery routine shall execute upon fault detection and will either allow the unit to continue running or to stop execution, depending on user setting.

3.4.5.6 Manual forcing. The Compact, Small Modular, and Large Modular programmable controller shall have the ability to manually force hardwired analog and discrete inputs and outputs. Removal of these forced I/O points shall be accomplished individually or entirely through the programming interface. The programming interface shall have the ability to display all active forces. The front panel of the programmable controller shall indicate whether a force is active with the use of a light, such as an LED, or other indicator.

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3.4.5.7 Indicators. The Compact, Small Modular, and Large Modular programmable controller shall include the following visual indicators:

a. Current operation mode

b. Current fault status

c. I/O forces present/active

d. Data transfer activity/status for any communication protocol in use

e. I/O module channel state

3.4.6 I/O capabilities.

3.4.6.1 Programmable controller I/O points. Each programmable controller shall be capable of addressing a minimum number of I/O points.

a. The Compact programmable controller shall be capable of addressing a minimum of 100 I/O points.

b. The Small Modular programmable controller shall be capable of addressing a minimum of 500 I/O points.

c. The Large Modular programmable controller shall be capable of addressing a minimum of 1,000 I/O points.

3.4.6.2 I/O modules. The quantities and types of I/O modules shall be provided with each PLC as specified (see 6.2). Programmable controllers shall have the capability to utilize input data modules with open wire diagnostic detection, which shall generate alerts, if so equipped, to the supervisory network.

3.4.6.3 I/O module data types. The Small Modular and Large Modular programmable controller module-defined data types shall include I/O module-specific information and diagnostic status for each module (see 6.3).

3.4.7 Communications.

3.4.7.1 Communications capabilities. The Compact, Small Modular, and Large Modular programmable controller shall contain communications capabilities in accordance with IEC 61131-5.

3.4.7.2 Ethernet networking port. The Compact, Small Modular, and Large Modular programmable controller shall have a dedicated Ethernet networking port, either embedded or able to be added via expandable communication modules that support networking and peer-to-peer messaging.

3.4.7.3 Networking media. Networking media shall have the ability to:

a. Communicate through copper media with dedicated RJ-45 (Registered Jack) copper connector type capability.

b. Communicate through fiber-optic media with dedicated LC fiber-optic connector type capability.

c. Utilize fiber-optic media converters as specified (see 6.2).

3.4.7.4 Networking topologies. The Small Modular and Large Modular programmable controller network shall support all standard industry configurations including any combination of bus, star, and ring.

3.4.7.5 Dual-homing. The Large Modular programmable controller shall have dual-homing communication capability.

3.4.7.6 Loss of communications. Upon loss of communications, the Compact, Small Modular, and Large Modular programmable controller shall have the following capabilities:

a. Individual programmable controllers shall continue to run and remain in a safe state of control.

b. Loss of communications between programmable controllers shall have the ability to configure to fail low, fail high, or to fail in last known state.

c. Control signals shall be configurable to fail last, fail low, or fail high, depending upon I/O module capability.

d. An I/O communication failure shall occur if communications to the module is lost.

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3.4.7.7 PLC processor-to/from-PLC processor (peer-to-peer) communications. The Small Modular and Large Modular programmable controller shall have the ability to send and receive data from other programmable controllers through the User Datagram Protocol (UDP) and Internet Protocol (IP) multicast.

3.4.7.8 Distributed I/O communications. The Small Modular and Large Modular programmable controller shall have the ability to send and receive data from distributed I/O devices via the network.

3.4.7.9 PLC processor-to/from-human machine interface (HMI) communications. The Small Modular and Large Modular programmable controller shall be capable of transferring data via Transmission Control Protocol/Internet Protocol (TCP/IP) and UDP.

3.4.7.10 Network bridging. The Small Modular and Large Modular programmable controller shall be capable of bridging to other types of networks (for example, Profibus).

3.4.7.11 Other communications protocols. The Compact, Small Modular, and Large Modular programmable controller shall support the following network protocols:

a. RS-232

b. RS-485

c. RS-422

d. Modbus RTU

e. ASCII

f. Profibus

g. Profinet

h. Ethernet/IP

i. ModbusTCP

3.4.8 Electrical service.

3.4.8.1 Power. The Compact, Small Modular, and Large Modular programmable logic controller shall operate in compliance with an electrical service of either of the following, as specified (see 6.2).

a. 120 volts alternating current (VAC) (85-264 VAC operating) single-phase, in the frequency range from 47 to 63 hertz in accordance with IEC 61131-2.

b. 24 volts direct current (VDC) (19.6-28.8 VDC operating) in accordance with IEC 61131-2.

3.4.8.2 Power supply. The power supply shall have the ability to supply regulated power to the programmable controller, backplane, and required I/O modules.

a. The Compact programmable controller shall distribute support power via internal connections to the integrated I/O.

b. The Small Modular and Large Modular programmable controller shall distribute support power to the local I/O/expansion modules via connecting cabling or through the backplane.

3.4.8.3 Power-up. Operation of the programmable controller shall be inhibited during power-up.

3.4.8.4 Fuse protection. The power supply shall have fuse protection downstream to prevent damage to the power supply in the event of overcurrent in accordance with MIL-DTL-2212.

3.4.8.5 Fluctuation. The power supply shall not fluctuate for loads between 10 percent and 100 percent of rated capacity.

3.4.8.6 Power loss. The Compact, Small Modular, and Large Modular PLC shall shut down upon a total loss of supply voltage. Control signals associated with that PLC shall be able to be programmed to fail low, fail high, or fail last.

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3.4.9 Program creation.

3.4.9.1 Programming software. Programming software shall be provided to interface with the Compact, Small Modular, and Large Modular programmable controller.

3.4.9.2 Programming software interface. The programming software interface shall connect to the Compact, Small Modular, and Large Modular programmable controller via an Ethernet interface or local CPU port.

3.4.9.3 Programming languages.

a. The Compact programmable controller shall be able to utilize Ladder (LD) and Function Diagram (FBD) in accordance with IEC 61131-3.

b. The Small Modular and Large Modular programmable controller shall have the capability to utilize LD, FBD, textual-programming languages, and graphic-programming languages in accordance with IEC 61131-3.

3.4.9.4 Program execution. The Compact, Small Modular, and Large Modular programmable controller shall have the ability to use continuous or periodic tasks for executing program logic, background processing of other processor functions, and processing of I/O information in accordance with IEC 61131-3.

3.4.9.5 Sub-system. The Small Modular and Large Modular programmable controller sub-system shall have the ability to be configured to interface with new I/O modules.

3.4.9.6 Data types and storage. The Small Modular and Large Modular programmable controller shall have the ability to store elementary data types as defined in IEC 61131-3.

3.4.9.7 Addressing descriptions. The Small Modular and Large Modular programmable controller shall have the ability to accept descriptions, comments, and symbols as part of non-executable programming lines.

3.4.9.8 Debugging. The Compact, Small Modular, and Large Modular programmable controller shall have the ability to facilitate debugging in accordance with IEC 61131-1.

3.4.9.9 Event detection programming. The Small Modular and Large Modular programmable controller shall have the ability to use conditional event detection programming. Output instructions shall include one-shot instructions that shall provide the capability of being triggered on either low-to-high (rising) or high-to-low (falling) rung conditions.

3.4.9.10 Interrupt function. The Compact, Small Modular, and Large Modular programmable controller interrupt mechanism of periodic tasks shall adhere to the IEC 61131-3 definition of pre-emptive multitasking.

3.4.9.11 Watchdog timers. The Small Modular and Large Modular programmable controller shall have the ability to use a watchdog timer that is user-defined and unique to that task.

3.4.9.12 Array configuration. The Small Modular and Large Modular programmable controller shall have the ability of using arrays configurable up to, but not limited to, three dimensions.

a. Array element manipulation instructions. Array element manipulation instructions such as high-speed “array copy” and “array fill”, “array to array” move, “element to array” move, “array to element” move, and “first in-first out” shall be supported. The four-function math instructions; instructions for performing “logical OR”, “logical AND”, and “exclusive OR”; and comparison instructions such as “less than”, “greater than”, and “equal to” shall be included. All instructions shall be capable of executing on both single words and arrays.

3.4.9.13 Status of I/O channels. The Small Modular and Large Modular programmable controller shall be capable of querying the current status of all channels via tags without any programming.

3.4.9.14 Latch functions. Latch functions shall be internal and programmable.

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3.4.9.15 Software timers and counters. The Compact, Small Modular, and Large Modular programmable controller shall have the ability to address software timers and counters.

a. Instructions shall permit programming timers in the on or off delay modes.

b. Timer programming shall also include the ability to interrupt timing without resetting the timers.

c. Counters shall be programmable using up-increment and down-increment.

3.4.9.16 Math functions. The Small Modular and Large Modular programmable controller shall support general math, logarithmic, and trigonometric functions in accordance with IEC 61131-3.

3.4.9.17 Diagnostic event. The Compact, Small Modular, and Large Modular programmable controller shall support analog and discrete I/O signal alarming with the following capabilities:

a. Ability to display an out-of-range high or out-of-range low alarm indication (see 6.4.8 and 6.4.9).

b. Analog alarm indications shall have an automatic or manual reset as specified (see 6.2).

c. Analog and digital I/O signals shall be fail-safe as specified (see 6.2).

3.5 Environmental conditions.

3.5.1 Operating temperature. The Compact, Small Modular, and Large Modular PLC shall operate at an ambient temperature (see 6.4.1) of 0 to 60 °C and a storage ambient temperature of -40 to 85 °C.

3.5.2 Cooling. The Compact, Small Modular, and Large Modular programmable controller, main and expansion chassis, and I/O modules shall provide free airflow convection cooling. No internal fans or other means of cooling, except heat sinks, shall be permitted.

3.5.3 Humidity range. The Compact, Small Modular, and Large Modular PLC shall operate in the relative humidity range of 5 to 95 percent non-condensation.

3.5.4 Inclination. The Compact, Small Modular, and Large Modular programmable controller shall operate when permanently inclined in any direction at any angle up to 15 degrees from the normal operating position and during temporary inclinations of up to 45 degrees in any direction from the normal operating position.

3.5.5 Electromagnetic interference (EMI). If the Compact, Small Modular, and Large Modular PLC is to be a component of a larger equipment or subsystem, the PLC shall be of sufficient electromagnetic capability design such that the PLC shall not inhibit or degrade the larger equipment's or subsystem's ability to meet the requirements of MIL-STD-461. If the PLC is to be used as a stand-alone unit not in an enclosure, the PLC itself shall be in accordance with MIL-STD-461.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the environmental test conditions specified in 3.5.1, 3.5.3, and 3.5.4.

4.3 First article inspection. First article inspection shall be performed on one to five samples of components of each size of programmable controller when a first article sample is required (see 3.1). This inspection shall include the examination of 4.5 and the tests of table I and 4.6 through 4.6.8.

4.4 Conformance inspection. Conformance inspection shall include the examination of 4.5 and the tests of table I and 4.6 through 4.6.8.

4.4.1 Sequence of inspections. Unless otherwise specified (see 6.2), the tests for overall examination, general operation, and testing of cybersecurity measures shall precede the functional testing of 4.6.1 through 4.6.8.

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TABLE I. First article tests and conformance inspections.

Inspection	Requirement(s)	First Article	Conformance
Examination and general operation	3.1 through 3.2.1	4.5	4.5
Cybersecurity	3.3	4.5	4.5
Performance	3.4	4.5	4.5
Duty	3.4.3	4.5	4.5
Diagnostics	3.4.4	4.5	4.5
CPU	3.4.5, 3.4.5.1 through 3.4.5.7	4.6.1	4.6.1
I/O	3.4.6, 3.4.6.1 through 3.4.6.3	4.6.2	4.6.2
Communications	3.4.7, 3.4.7.1 through 3.4.7.11	4.6.3	4.6.3
Power	3.4.8, 3.4.8.1 through 3.4.8.6	4.6.4	4.6.4
Program creation	3.4.9, 3.4.9.1 through 3.4.9.17	4.6.5	4.6.5
Operating temperature	3.5.1 through 3.5.2	4.6.6	4.6.6
Humidity	3.5.3	4.6.6	4.6.6
Reliability and maintainability	3.4.1 through 3.4.2	4.6.7	4.6.7
Inclination	3.5.4	4.6.8	4.6.8

4.5 Examination, general operation, and cybersecurity. Equipment shall be examined and subjected to a performance test to determine compliance with the requirements specified in 3.1 through 3.5.4. Non-compliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. These examinations shall not require disassembling the unit in such a manner that its performance, durability, or appearance would be affected. The general operation shall include energizing and operating the equipment to ascertain that the required operating features of the unit function properly, as well as demonstration of the capability and operability of the cybersecurity measures of 3.3.

4.6 Tests. The tests listed below shall be performed on the Compact, Small Modular, and Large Modular programmable controller. All tests shall be conducted at an ambient temperature range between 10 and 40 °C (50 and 104 °F). Equipment shall be operated at nominal voltage and frequency during the tests. When practicable, tests shall be performed without interruption.

4.6.1 CPU.

a. Test and demonstrate all configuration types, functions, and operation modes of which the device is capable to determine conformance to 3.2.1, 3.4.5.1, and 3.4.5.4.

b. Demonstrate the ability to display the firmware version locally or via network interface, if applicable, to determine conformance to 3.4.5.2.

c. Verify that a PLC program can be uploaded, stored, and executed in system memory.

d. Conduct memory back-up testing in accordance with IEC 61131-2 to determine conformance to 3.4.5.3.

e. Demonstrate minor and major faults in the processor and verify the ability to create a force (see 6.4.6) in the programmable controller software logic. Verify that the appropriate fault indications are activated and that the appropriate indicators are illuminated to determine conformance to 3.4.5.5, 3.4.5.6, and 3.4.5.7.

f. Test and demonstrate that all other visual indicators on the front panel of the programmable controller function as specified in 3.4.5.7.

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4.6.2 I/O points.

a. Verify accurate count of I/O points to determine conformance to 3.4.6.1.

b. Test and demonstrate the proper operation of both embedded and distributed I/O modules, stimulating the appropriate voltage and current signals to demonstrate conformance to 3.4.6.2 and 3.4.6.3.

4.6.3 Communications.

a. Test and demonstrate all executable commands and status requests for all installed network interface protocols to determine conformance to 3.4.7.

b. Verify that the Ethernet networking port is installed and functional as specified in 3.4.7.2.

c. Test and demonstrate the networking media functionality for select programmable controllers to determine conformance to 3.4.7.3.

d. Test and demonstrate the networking functionality of select programmable controllers to determine conformance to 3.4.7.4, 3.4.7.5, and 3.4.7.7 through 3.4.7.10.

e. Test and demonstrate the communication loss requirements of 3.4.7.6.a through 3.4.7.6.d.

f. Verify all sizes of programmable controller are compatible with the protocols of 3.4.7.11.

4.6.4 Electrical service conditions. Programmable controller power shall be tested in accordance with MIL-DTL-2212 for steady state voltage and frequency, transient voltage and frequency, spike voltage, power interruption, leakage current, output leakage current, and output voltage drop to determine conformance to 3.4.8.

4.6.5 Program creation. Utilizing IEC 61131-3, the programming software functionality for select programmable controllers shall be tested and demonstrated to determine conformance to 3.4.9.

4.6.6 Temperature and humidity. For conformance, this inspection shall be performed on the first production run only. Temperature and humidity tests are not required on subsequent production runs unless a change is made in design. Temperature testing shall be conducted in accordance with MIL-STD-810, Method 501 (high temperature) and Method 502 (low temperature). For high temperature, the specific test shall be “Basic Hot” for three cycles. For low temperature, the specific test shall be “Basic Cold” with the temperature range for the test relaxed to -2 to 0 °C. Humidity testing shall be conducted in accordance with MIL-STD-810, Method 507, cycle B1 for seven cycles. The PLC shall be tested for conformance to the relative humidity requirements specified in 3.5.3.

4.6.7 Reliability and maintainability. The controller shall be analyzed to verify the requirements for reliability specified in 3.4.1. The validation procedures in MIL-HDBK-217 may be used. The controller shall be demonstrated to meet the requirements for maintenance and repair specified in 3.4.2. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during conformance testing. The demonstration procedures in MIL-HDBK-470 may be used.

4.6.8 Inclination. Equipment shall pass a general operation test when inclined from bulkhead mounted orientation to the maximum angles specified in 3.5.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The programmable controllers covered by this specification are intended for use on naval surface ships and submarines. The controllers procured to this specification are form, fit, and function replacements for prior revisions. The Government will be responsible for scanning all software program updates prior to upgrading the PLC to detect any malicious software. The Government will be responsible for verifying that any alterations are in accordance with the intended use of the PLC. Table II should be used to select the most appropriate PLC type for a given application.

TABLE II. Decision criteria: high-level decisions to select PLC type.

Characteristic	Compact PLC	Modular PLC – Small	Modular PLC – Large
Processing Power	< = 32-bit processor	< = 32-bit processor	> = 32-bit processor
Memory	< = 1MB	> = 1MB user memory	> = 2MB user memory
Communication Redundancy	No	No	Yes
Single or Multiple System Control	Single	Single, Multiple	Multiple
Single Location or Distributed I/O	Single	Single	Distributed
I/O Size	Minimum of 100	Minimum of 500	Minimum of 1,000
I/O Diversity	Discrete and some analog	Discrete and some analog	All types of discrete and analog
Interconnectivity	Serial, Ethernet	Serial, Ethernet	Serial, Ethernet, Distributed I/O
Programming Software Logging Capability	No	Yes	Yes
Physical Space Limitation	Size of self-contained unit.	Size of the programmable controller itself plus expansion modules.	Size of the programmable controller itself plus chassis and expansion modules.
Programming Requirements	LD, FBD	LD, FBD, SFC, SCL, SL	LD, FBD, SFC, SCL, SL

NOTES:
1. LD = Ladder
2. FBD = Function Block Diagram
3. SFC = Sequential Function Chart
4. SCL = Structured Control Language
5. SL = Statement List

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article inspection is required (see 3.1).
c. Requirements for base processor memory (see 3.4.5.3).
d. Quantities and types of I/O modules (see 3.4.6.2 and 6.3).

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e. Fiber-optic media converters for networking media (see 3.4.7.3.c).

f. Power (see 3.4.8.1).

g. Automatic or manual alarm resets (see 3.4.9.17.b).

h. Analog and digital I/O signal fail-safe (see 3.4.9.17.c).

i. Inspection conditions, if other than those specified in 3.5.1, 3.5.3, and 3.5.4 (see 4.2).

j. Sequence of inspections (see 4.4.1).

k. Packaging requirements (see 5.1).

6.3 I/O module capabilities. The Compact, Small Modular, and Large Modular PLC should support the following input/output module types:

a. 10-30 VDC Discrete Input: Capable of being selectable to reset or hold last state during a fault. Capable of detecting normally open or normally closed contacts.

b. 0-30 VDC Diagnostic Discrete Input: Capable of detecting a wire break versus an open contact.

c. Resistance Temperature Detector (RTD) Input: Capable of reporting °C, °F, or current for 100-ohm platinum, 120-ohm nickel, or 10-ohm copper sensors.

d. Thermocouple/Millivolt (mV) Input: Capable of interfacing with type J and K thermocouples.

e. Analog Input: Capable of 0-5 VDC, ±5 VDC, 4-20 milliampere (mA), 0-20 mA. Capable of supporting 1,000-volt insulation.

f. Isolated Analog Input: Capable of 0-5 VDC, ±5 VDC, 0-10 VDC, ±10 VDC, 4-20 mA, 0-20 mA.

g. Analog Output: Capable of 0-5 VDC, ±5 VDC, 4-20 mA, 0-20 mA. Capable of supporting 1,000-volt insulation.

h. Isolated Analog Output: Capable of 0-5 VDC, ±5 VDC, 0-10 VDC, ±10 VDC, 4-20 mA, 0-20 mA.

i. Isolated Discrete Input: Capable of 0-5 VDC, ±5 VDC, 24 VDC.

j. Isolated Discrete Output: Capable of 0-5 VDC, ±5 VDC, 24 VDC.

k. Relay Discrete Output.

l. 230-VAC Individually Isolated Output.

m. 10- to 30-VDC Discrete Output.

n. 30- to 60-VDC Discrete Output.

o. 48- to 125-VDC Discrete Output.

p. 30- to 60-VDC Discrete Input.

q. 48- to 125-VDC Discrete Input.

r. 24-VDC Discrete Input.

s. 10- to 30-VAC Discrete Input.

t. 10- to 30-VAC Discrete Diagnostic Input: Capable of detecting a wire loss.

u. 79- to 132-VAC Discrete Input.

v. 79- to 132-VAC Diagnostic Digital Input: Capable of detecting a wire loss.

w. 10- to 30-VAC Diagnostic Discrete Output capable of detecting a loss of load or output power.

x. 79- to 132-VAC Discrete Output.

y. 79- to 132-VAC Diagnostic Discrete Output: Capable of detecting a loss of load or output power.

z. 10- to 30-VDC Diagnostic Discrete Output: Capable of detecting a loss of load or output power.

aa. 9- to 32-VAC Isolated AC Input.

bb. 79- to 132-VAC Input.

cc. Differential/Single-ended Voltage/Current Analog Input.

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dd. Individually Isolated Differential Voltage/Current Analog Input.

ee. Differential/Single-Ended Input/Output Analog.

ff. Single-Ended Current Analog Input.

gg. Single-Ended Voltage Analog Input.

hh. 240-VAC Input.

ii. 24-VDC Sink/Source Input.

jj. AC/DC Relay Output.

kk. 100/240-VAC Solid-State Output.

ll. 24-VDC Solid-State Output.

mm. Voltage/Current Analog Output.

nn. Individually Isolated Voltage Output.

oo. Output Discrete can be Continuous or Momentary outputs.

pp. Frequency Analog Input.

qq. Frequency Analog Output.

6.4 Definitions.

6.4.1 Ambient temperature. The temperature of the medium, such as air, water, or earth, into which the heat of the equipment is dissipated. For self-ventilated equipment, the ambient temperature is the average temperature of the air in the immediate neighborhood of the equipment. For air or gas-cooled equipment with forced ventilation or secondary water-cooling, the ambient temperature is taken as that of the in-going air or cooling gas. For self-ventilated enclosed (including oil-immersed) equipment considered as a complete unit, the ambient temperature is the average temperature of the air outside of the enclosure in the immediate neighborhood of the equipment.

6.4.2 Continuous duty. Continuous duty is a requirement of service that demands operation at a substantially constant load for an indefinitely long time.

6.4.3 Fail high. A state or condition where a device's control variable/position increases to the maximum range value or higher upon loss of power or loss of communications.

6.4.4 Fail last. A state or condition where a device remains at its current control variable/position upon loss of power or loss of communications.

6.4.5 Fail low. A state or condition where a device's control variable/position decreases to the minimum range value or lower upon loss of power or loss of communications.

6.4.6 Forcing. A function of the programmable controller's programming software in which an analog or discrete I/O data point is manually modified to a desired value differing from the live value from the field.

6.4.7 Input/output (I/O) module. Input devices accept signals from the process elements and translate them into signals that can be used within the controller. Output devices convert controller signals into external signals used to control the process.

6.4.8 Out-of-range high. When the signal is greater than the adjustable or pre-configured range.

6.4.9 Out-of-range low. When the signal is less than the adjustable or pre-configured range.

6.4.10 Programmable logic controller (PLC). A digitally operating electronic apparatus that uses a programmable memory for the internal storage of instructions for implementing specific functions such as logic, sequencing, timing, and counting.

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6.4.11 Service of a controller. The specific application in which the controller is to be used, for example:

a. General purpose

b. Definite purpose:

(1) Crane and hoist

(2) Elevators

(3) Machine tool

6.4.12 Supervisory network/system. A network or system consisting of analog and digital electronic devices (possibly computers) containing output signals that either directly or indirectly effect change external to the system and control a physical process. This includes systems designated as Machinery Control Systems; supervisory control and data acquisition systems; warfare, navigation, and hull, mechanical, and electrical systems; and industrial control systems. Information technology systems are excluded. (Definition per: SCG ID# 10-040, U.S. Navy Afloat Control Systems Cyber Security Classification Guide [SCG].)

6.5 Subject term (key word) listing.

Module, input, output

PLC

Processor

Programmable logic controller

6.6 Equipment modifications. Any redesign or modification of the contractor's standard product to comply with specified requirements or any necessary redesign or modification following failure to meet the specified requirements will receive particular attention for adequacy and suitability.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85

Review activities:
Army – MI
Navy – AS, CG
Air Force – 03, 19
DLA – CC

Preparing activity:
Navy – SH
(Project 5998-2019-029)

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